Last time

$H = \text{span}\{v_1, \dots, v_p\}$

Check whether $\{v_1, \dots, v_p\}$ is lin. dep.

If $v_k = d_1 v_1 + \dots + d_{k-1} v_{k-1}$ throw out v_k.

Then $H = \text{span}\{v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_p\}$

Repeat with smaller set until you get a basis for H.

In summary, a spanning set contains a basis.

Example

Let $H = \text{Span}\left\{ \underset{v_1}{\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}, \underset{v_3}{\begin{bmatrix} 0 & 0 \\ -2 & -1 \end{bmatrix}} \right\} \subseteq M_{2\times 2}$

Solve $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0 \iff \begin{bmatrix} 2c_1 + c_2 & 0 \\ c_2 - 2c_3 & c_1 + c_2 - c_3 \end{bmatrix} = 0$

$$\iff \begin{cases} 2c_1 + 1c_2 + 0c_3 = 0 \\ 0c_1 + 1c_2 - 2c_3 = 0 \\ 1c_1 + 1c_2 - 1c_3 = 0 \end{cases}$$

To solve this we row reduce $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$

to obtain $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$.

Solution set is $\left\{ c_3 \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, c_3 \in \mathbb{R} \right\}$

Take $c_3 = 1$ the $\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ is a solution

$\Rightarrow \quad -1 v_1 + 2 v_2 + 1 v_3 = 0$

$\Rightarrow \quad \{v_1, v_2, v_3\}$ is lin. dep

and $\quad v_3 = -2 v_2 + v_1$

Throw out v_3 to get $\text{Span}\{v_1, v_2\} = H$.

Since $v_2 \neq c v_1$ for any $c \in \mathbb{R}$, $\{v_1, v_2\}$ is lin. ind. and $\{v_1, v_2\}$ is a basis for H

Example

Take $H = \text{Nul}(A)$ for $A \in M_{m \times n}$.

$\text{Nul}(A)$ = solution set of $Ax = 0$

We can always write $\text{Nul}(A) = \text{span}\{v_1, \dots, v_k\}$.

for some $v_1, \dots, v_k \in \mathbb{R}^n$, where k is the number of free variables. (Exercise 4.2 3-6)

Theorem: $\{v_1, \dots, v_k\}$ is a basis for $\text{Nul}(A)$

Example

Take $H = \text{Col}(A)$ for some $A \in M_{m \times n}$

Then $\text{Col}(A) = \text{span}\{a_1, \dots, a_n\}$ where

$A = [a_1 \cdots a_n]$ and $a_1, \dots, a_n \in \mathbb{R}^m$

Thm 6

Let B be the REF of A and $1 \le j_1, \dots, j_k \le n$ be the column numbers with pivots (leading ones).

Then $\{a_{j_1}, \dots, a_{j_k}\}$ is a basis for $Col(A)$

Example

Take $A = [a_1 \cdots a_n]$ and suppose

REF of A is $B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \in M_{4\times 5}$

$$\begin{array}{ccccc} \uparrow & & \uparrow & & \uparrow \\ 1 & & 3 & & 5 \end{array}$$

Thm 6 tells us that $Col(A)$ has basis

$\{a_1, a_3, a_5\}$

4.4 Coordinate systems

In this section V is a (real) vector space.

Idea: Given a basis $\mathcal{B} = \{b_1, \dots, b_n\}$ for V.
Convert information about V into information about $\mathbb{R}^n$.

Theorem 8 (Unique representation Theorem)

Suppose $\mathcal{B} = \{b_1, \dots, b_n\}$ is a basis for V.

Then every $v \in V$ can be written as

$$v = c_1 b_1 + \cdots + c_n b_n$$

for a <u>unique</u> choice of scalars $c_1, \ldots, c_n \in \mathbb{R}$.

Def. Suppose $\mathcal{B} = \{b_1, \ldots, b_n\}$ is a basis for V. Then the <u>coordinate vector of $v \in V$</u> (relative to $\mathcal{B}$) is

$$[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n.$$

where $v = c_1 b_1 + \cdots + c_n b_n$ is as in Thm 8.

Example

Take $V = \mathbb{P}_1$ and $\mathcal{B} = \{1, x\}$.

Let $p(x) = a_0 1 + a_1 x$ where $a_0, a_1 \in \mathbb{R}$

Then $p \in \mathbb{P}_1$ and $[p]_{\mathcal{B}} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$

e.g. if $p(x) = 2 + 3x$

Then $[p]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Example

Take $V = \mathbb{R}^2$ and $\mathcal{B} = \left\{ \underset{b_1}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}, \underset{b_2}{\begin{bmatrix} 1 \\ 2 \end{bmatrix}} \right\}$

(Take for granted that $\mathcal{B}$ is a basis).

Take $v = \begin{bmatrix} 1 \\ 6 \end{bmatrix} \in \mathbb{R}^2$. Want to write

$$v = c_1 b_1 + c_2 b_2$$

$$\iff \begin{bmatrix} 1 \\ 6 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\iff \begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ 2c_2 \end{bmatrix}$$

$$\iff c_2 = 3, \quad c_1 = -2$$

$$\iff v = -2b_1 + 3b_2$$

$$\iff [v]_B = \begin{bmatrix} 1 \\ 6 \end{bmatrix}_B = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Remarks

Take $V = \mathbb{R}^n$ and $B = \{b_1, \dots, b_n\}$. For any

$$v = c_1 b_1 + \cdots + c_n b_n = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad (*)$$

By def'n of matrix multiplication

e.g in above example $\begin{bmatrix} b_1 & b_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

and $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix} = v$.

Define $P_B = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix}$ as above.

P_B = <u>change of coordinate matrix</u>

(From B to the standard basis $\{e_1, \dots, e_n\}$)

It follows from (*) that $v = P_B [v]_B$.

Since the columns of P_B are linearly ind.

P_B is invertible and so $P_B^{-1} v = \cancel{P_B^{-1}} \cancel{P_B} [v]_B$

$= [v]_B$.

$$\Rightarrow \quad P_B^{-1} v = [v]_B.$$

Example

Take $V = \mathbb{R}^2$, $b_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and

$B = \{b_1, b_2\}$ (a basis).

Then $P_B = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ and

$$P_B^{-1} = \frac{1}{(2)(1)-(1)(-1)} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}.$$

and $$\begin{bmatrix} 4 \\ 5 \end{bmatrix}_B = P_B^{-1} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 9 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Remarks

Suppose $B = \{b_1, \dots, b_n\}$ is a basis for V.

We can define $T: V \to \mathbb{R}^n$ and $S: \mathbb{R}^n \to V$

as follows

$$T(v) = [v]_B \qquad \forall v \in V$$

$$S\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right) = a_1 b_1 + \dots + a_n b_n \qquad \forall \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n.$$

Theorem 9 (Paraphrase)

T and S are linear transformations. Moreover $T^{-1} = S$ i.e. $S(T(v)) = v$ and $T\left(S\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right)\right) = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.

Def. An invertible linear transformation is called an <u>isomorphism</u>. Two vector spaces with an isomorphism between them are <u>isomorphic</u>.

Umbrella Thm: Two isomorphic vector space have identical properties as vector spaces.

4.5. The dimension of a vector space

In this section $B = \{b_1, \ldots, b_n\}$ is a basis for a (real) vector space V.

Def. The <u>dimension</u> of $V = \dim V = n$.
(the number of vectors in a basis)

Theorem 11

Every basis for V has the same number of vectors.

Theorem 10

If $\{v_1, \ldots, v_p\} \subseteq V$ and $p > n = \dim V$

then $\{v_1, \ldots, v_p\}$ is linearly dependent.

Example

Recall that $\{1, x, x^2\}$ is a basis for $\mathbb{P}_2$.

$\Rightarrow \dim \mathbb{P}_2 = 3$.

If $p_1(x) = 1+x+x^2$, $p_2(x) = 2+3x^2$, $p_3(x) = 2-x+x^2$

$p_4(x) = x - x^2$. the $\{p_1, p_2, p_3, p_4\}$ is a lin. dep. subset of $\mathbb{P}_2$.

Consequence of Thm 10 is that lin. ind. subsets have size $\dim V$ or less.

Theorem 13

① Any lin. ind. set $\{v_1, \dots, v_n\} \subseteq V$ where $n = \dim V$ is a basis. (so $\operatorname{span}\{v_1, \dots, v_n\} = V$.)

② If $\operatorname{span}\{v_1, \dots, v_n\} = V$ where $n = \dim V$ the $\{v_1, \dots, v_n\}$ is a basis (so is lin. ind.)